

PALLETPILOT · REQUIREMENTS

PalletPilot requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.5	2026-09-26	Amish Chadha	Budget approved by Amish
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	Record decisions PLP-DDR-001 (R7 redefined for the lidar layer, R17 to \$1,550, 1,500 kg on level floors); status from PLP-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-24	Amish Chadha	Initial scaffold

PalletPilot requirements

These requirements are checked by calculation in PLP-CAL-001. Eleven of eighteen requirement lines are met on paper, two are not met (R5 bearing accuracy, R15 kit mass), two are at risk (R7, R9), and three cannot be verified at TRL 3 (R1, R6, R16). Targets are still to be validated with users after site visits (see PLP-PRB-001). Decisions recorded in PLP-DDR-001 on 2026-09-25 changed R2, R4, R7 and R17; decisions in PLP-DDR-002 on the same date restated R4 and R8 (bumper-only and untested follow mode at 0.15 m/s) and R9 (two contactors in series). On 2026-09-26 Amish approved the budget: R17's target is \$1,610, which covers the priced BOM, so R17 is met (PLP-DDR-002).

The **design load case** is a 1,000 kg pallet on a 64 kg donor jack with a kit of about 42.0 kg, about 1,106 kg in total, on a level, dry, sealed concrete floor.

The **design duty** is one 8 h shift of 60 pallet moves, each 40 m loaded and 40 m empty, with four starts in each direction.

Table 1. Requirements.

ID	Requirement	Target	Verification	Status (PLP-CAL-001)
R1	Fit a common manual jack	Bolts to 27 x 48 in (685 x 1220 mm), 2,500 kg manual jacks with a common steering yoke; no welding, cutting or drilling of load-bearing parts; fitted in 2 h or less and removed in 1 h or less with hand tools	Survey of three donor models; fitting sequence review	Not verifiable at TRL 3; the model assumes one donor geometry
R2	Move the design load	Start and move 1,000 kg on a level floor; kit rating 1,500 kg at 0.8 m/s or less on level floors, below the donor rating (decided, PLP-DDR-001 D7)	Traction and torque calculation	Met: 25.5 N·m per motor needed of 30 N·m specified; 750 N traction. A 1,500 kg ramp start is outside the rating
R3	Handle short ramps	Start, climb and hold the design load on a 2 % grade	Traction, torque and brake calculation	Met: start needs 488 N of 750 N (margin 1.54, 1.23 at friction 0.4)
R4	Walking-pace speeds	Walk mode 1.2 m/s (4.3 km/h) or less handle-end first and 0.8 m/s or less forks first; creep 0.3 m/s or less with the handle near upright; follow mode 0.6 m/s or less, and 0.15 m/s or less until the lidar layer is built and tested (decided, PLP-DDR-001 D6; interim limit lowered from 0.2 m/s to the bumper-only limit by PLP-DDR-002)	Controller limits; later timed runs	Met by design (software limits)
R5	Follow the operator	In follow mode, hold a 1.5 m gap within ± 0.3 m and a bearing within $\pm 10^\circ$ (2σ) of the tag at walking pace	UWB error budget; later tracking trials	Not met: gap ± 0.06 m, but bearing about $\pm 16^\circ$ with a 450 mm anchor baseline and filtering
R6	Stop if the operator is lost	Stop command within 0.3 s of losing the tag link, a gap over 3 m, a tag stop press or the tag leaving a $\pm 45^\circ$ cone	Firmware design review; later bench test	Not verifiable at TRL 3; timing budget 0.20 s
R7	Stop before hitting a person in follow mode	The lidar layer detects a person in the travel path and the truck stops without contact from 0.6 m/s with the design load, with the bumper as the last layer (redefined for option 1A, D1)	Stopping calculation; later test with an ISO 3691-4 style test piece	At risk: stop 0.42 m level and 0.76 m worst case, inside a 0.86 m protective field; the sensor is not safety-rated and detection of dark clothing is unproven
R8	Stop on bumper contact	Bumper contact removes drive power and applies the brakes within 100 ms; any operation that relies on the bumper alone limited to 0.15 m/s or less (decided, PLP-DDR-002; was 0.2 m/s)	Stop chain design review	Met on paper: 0.10 s chain; 0.15 m/s stops in 38 mm of the 40 mm edge travel
R9	Emergency stop	Two hardwired emergency stops (tiller head and enclosure) plus the tag stop; the hardwired stops cut drive power through a dual-channel safety relay and two main contactors in series (decided, PLP-DDR-002), independent of software; target performance level PL d under ISO 13849-1	Stop chain review; PL calculation	At risk: dual-channel relay and two contactors in series; PL not yet calculated

ID	Requirement	Target	Verification	Status (PLP-CAL-001)
R10	Park safely	Brakes apply with power off and hold the design load on a 2 % grade	Brake torque check against datasheet	Met on paper: 40 N·m specified against 21.7 N·m needed; datasheet to confirm
R11	Keep the operator in control	Walk mode drives only with the handle between about 20° and 70° from vertical; belly-reverse paddle on the tiller head reverses the truck when pressed	Design review; model check	Met: the handle clears the enclosure by 40 mm at 70° and 14 mm when lowered flat
R12	Work a full shift	Complete the design duty on one charge with 20 % or more of usable energy left	Energy calculation	Met: 293 Wh used of 410 Wh usable (29 % left)
R13	Charge overnight	Full charge in 5 h or less from a 120 V or 230 V outlet through a certified charger	Charger datasheet	Met: 4.5 h at 5 A
R14	Push by hand when unpowered	One lever lifts the drive wheels clear in 10 s or less (brakes need no release once the wheels are clear); unpowered push force no more than 10 % above the bare jack	Design review; later push-force test	Met on paper: +4.0 % push force; lever effort about 117 N
R15	Keep the jack usable	Kit mass 40 kg or less; overall length increase 300 mm or less at floor level; lowered fork height and pallet entry unchanged	Mass estimate; model check	Partly met: 290 mm added length (met); 42.0 kg (not met , 2.0 kg over; accepted for now, PLP-DDR-002)
R16	Indoor environment	Operate at 0 to 40 °C; electronics and connectors IP54; no charging below 0 °C	Datasheets and design review	Not verifiable at TRL 3; set by the specification of bought parts
R17	Affordable	Kit parts cost \$1,610 or less, donor jack excluded (budget decided, D2; approved at \$1,610 by Amish, 2026-09-26, PLP-DDR-002)	Priced BOM (bom/bom.csv)	Met: \$1,610, at the budget with no margin; motor price still to be rechecked at a quote

■ Requirements not met or at risk

- **R5, follow bearing accuracy, is not met.** Filtering brings the error to about $\pm 16^\circ$; $\pm 10^\circ$ would need a 706 mm anchor baseline, wider than the jack. Amish decided to evaluate angle-of-arrival modules and lidar leg tracking at TRL 4, which is on hold (PLP-DDR-002).
- **R15 mass is not met** by 2.0 kg. The 30 N·m class hub motors weigh about 7 kg each; the second contactor adds 0.3 kg. Amish accepted the overrun for now, to be rechecked at the motor quote (PLP-DDR-002).
- **R17, cost, is met with no margin:** \$1,610 against the \$1,610 budget Amish approved on 2026-09-26 (it was \$60 over the former \$1,550). Any rise at the motor quote would take it over again (PLP-DDR-002).
- **R7 is at risk.** The stopping distances fit the lidar field on paper, but the sensor is not safety-rated, so this cannot support an ISO 3691-4 claim.
- **R9 is at risk** until the stop chain, now with two contactors in series, has a PL calculation.

■ Assumptions

- Rolling resistance 1.2 % and breakaway resistance 2.5 % for polyurethane steer and load wheels on smooth concrete. These are assumptions to be measured on a real jack.
- Tread-to-floor friction coefficient 0.5 for the drive wheels on dusty concrete, 0.4 as a low case; lower on wet or polished floors.
- The drive wheels carry 1.5 kN of spring preload and no share of the pallet load.
- Motor plus driver efficiency 70 %; controls, UWB, lidar and lights draw about 9.2 W in total.
- Stop delays of 0.25 s for the lidar layer and 0.10 s for the bumper and emergency stops.
- R8 is met by the 0.15 m/s speed limit, not by a longer-travel edge; a firmware or hardware fault that lets the truck exceed 0.15 m/s in a bumper-only mode would defeat it.
- The legal status of the retrofit (OSHA 1910.178(a)(4)) and of follow mode (ISO 3691-4) is open; targets R7 and R9 anticipate the stricter answer. The kit is documented for research use in a closed area (D8).